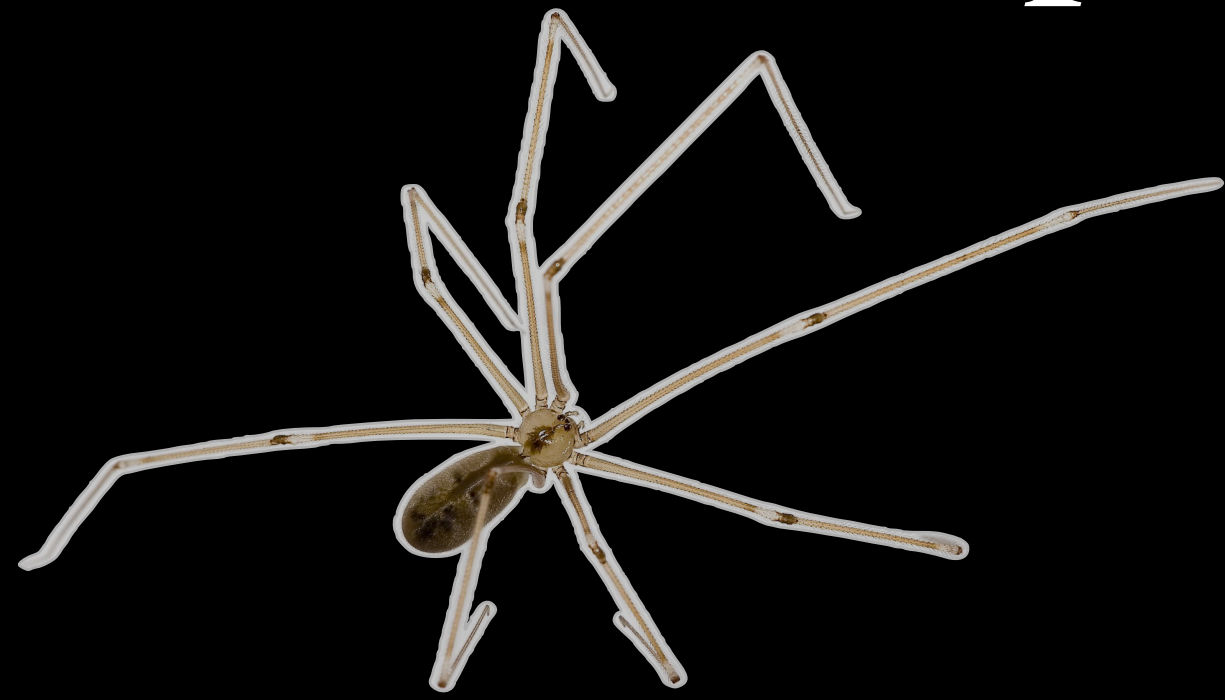




Pholcus phalangioides Response to Vibrational Frequencies



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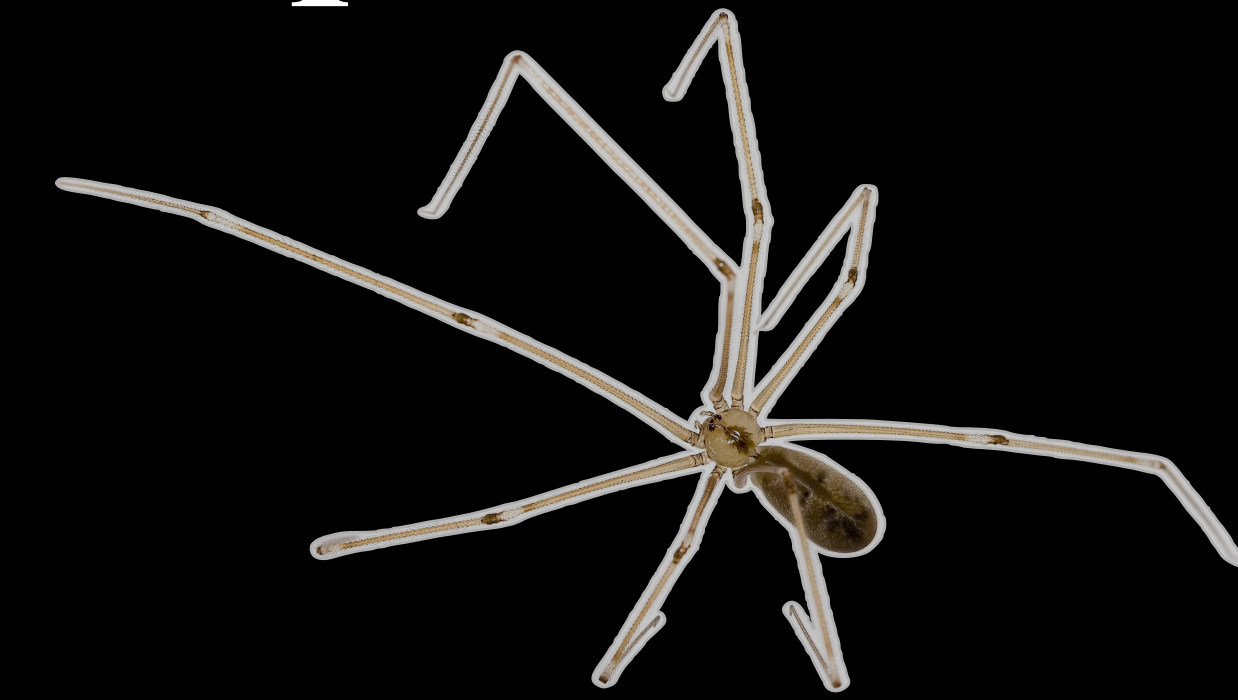


Figure 1: A *Pholcus phalangioides*, commonly referred to as the cellar spider or daddy-long-legs spider.

INTRODUCTION

Background

- Web-building spiders are known to have lyriform organs in their leg joints that allow them to sense vibrations [1].
- Prior research has shown that orb-weaving spiders use their webs to sense and respond to vibrations, and they are more sensitive to high frequency vibrations [2].
- Pholcus phalangioides* (Figure 1) live in webs, but are not orb weavers.

Research Question

- Do *Pholcus phalangioides* respond differently to high versus low-frequency vibrations on their webs?

Hypothesis

- Pholcus phalangioides* engage in defensive behaviors in the presence of low-frequency vibrations, and engage in predatory behaviors in the presence of high-frequency vibrations.

METHODS & MATERIALS

- 48 spiders of various sizes (small, medium, and large) were collected from buildings on the UMass Amherst campus.
- Each spider was placed in its own small arena (Figure 2) and left undisturbed for 5 days to build a web.
- The population was subdivided into 3 groups of 16 random specimens. Each group was tested with one frequency via tuning forks: 512 Hz, 341.3 Hz, or 256 Hz.
- To test the individuals, a vibrating tuning fork was put in contact with the web of each spider. Their reactions (reaction occurrence, time until reaction, duration of reaction, and reaction type - defensive, predatory, or reflexive) were recorded for each individual spider. Each individual was tested only once.
- Chi-squared test of independence and ANOVA statistical tests were used to analyze the data.

Table 1: Reaction type

Reaction Type:	Behaviors:
Defensive	Moving away from the fork, whirling (rapidly bobbing in the web)
Predatory	Approaching the tuning fork, touching the fork with its legs, wrapping the fork in web
Reflexive	Any brief movement in a direction other than towards or away from the tuning fork



Figure 2: Image of spider in its arena.

Table 2: Summary of statistical test results.

Comparison	Test Type	P-value	Significance
Size vs. reaction occurrence	Chi-Squared	0.0623	*
Size vs. defensive reaction	Chi-Squared	0.1534	NS
Size vs. predatory reaction	Chi-Squared	0.0573	*
Size vs. reflexive reaction	Chi-Squared	0.827	NS
Size vs. time until reaction	ANOVA	0.6537	NS
Size vs. reaction duration	ANOVA	0.3812	NS
Frequency vs. reaction occurrence	Chi-Squared	0.3499	NS
Frequency vs. defensive reaction	Chi-Squared	0.5194	NS
Frequency vs. predatory reaction	Chi-Squared	0.5488	NS
Frequency vs. reflexive reaction	Chi-Squared	0.5194	NS
Frequency vs. time until reaction	ANOVA	0.4887	NS
Frequency vs. reaction duration	ANOVA	0.3501	NS

Table 3: Size vs. reaction occurrence chi-square test of independence.

		Size			Row Total:
		Small	Medium	Large	
Reaction Occurrence?	Yes	15	17	8	40
	No	1	7	0	8
Column Total:		16	24	8	Grand Total: 48

Table 4: Size vs. predatory reaction chi-square test of independence.

		Size			Row Total:
		Small	Medium	Large	
Predatory Reaction?	Yes	9	5	2	16
	No	7	19	6	32
Column Total:		16	24	8	Grand Total: 48

RESULTS

- There were two findings of marginal significance (Table 1):
 - Size vs. reaction occurrence (chi-squared test of independence, $X^2 = 5.55$, $p = 0.0623$) (Table 2).
 - Size vs. predatory reaction (Chi-squared test of independence, $X^2 = 5.72$, $p = 0.0573$) (Table 3).
- No other findings were significant.

DISCUSSION

- The initial hypothesis of this experiment was rejected.
- The null hypothesis was accepted: *P. phalangioides* may not behave differently when presented with high vs. low vibrational frequencies in their webs.
- This may have been due to the imbalance in the number of spiders per size class in the sample. If the sample included a more equal portion of large individuals, the statistical tests relating to size vs. reaction type may have had significant results.
- Some error potential can be attributed to the unavoidable disturbance of the spiders prior to testing while opening the arenas, which could have caused behavioral changes in the specimens
- While the literature states that orb weaver spiders have different behaviors when they sense different frequencies [2], it appears that the same is not true for *Pholcus phalangioides*.
- Sensory perception in spiders varies among wandering vs. web-building spiders. It is now clear that perception also varies within the web-building spider species as well!
- Importance: emphasizes the variation in sensory perception within animal groups!

References

- [1] Seo et al. (2020) Lyriform vibration receptors in the web-building spider, *Nephila clavata* (Araneidae: Araneae; Arachnida). *Entomological Research*, 50: 586–593. [2] Mortimer, B. (2019). A Spider's Vibration Landscape: Adaptations to Promote Vibrational Information Transfer in Orb Webs. *Integrative and Comparative Biology*, 59(6), 1636–1645.